

Homework 1 with Solutions – Fall 2015

From Kurose-Ross - Ch. 1

Problem 2 (8 points)

Problem 6 (12 points)

Problem 10 (10 points)

Problem 12 (10 points)

Problem 13 (10 points)

P2. Equation 1.1 gives a formula for the end-to-end delay of sending one packet of length L over N links of transmission rate R . Generalize this formula for sending P such packets back-to-back over the N links.

Equation 1.1 says

$$d_{\text{end-to-end}} = N \frac{L}{R} \quad (1.1)$$

The packets are transmitted back-to-back, and that all links have the same transmission rate R . The first packet arrives after NL/R , and it takes $(P - 1) \cdot L/R$ more time to transmit the rest of them. Therefore the answer is

$$d = (N + P - 1) \cdot \frac{L}{R}$$

P6. This elementary problem begins to explore propagation delay and transmission delay, two central concepts in data networking. Consider two hosts, A and B, connected by a single link of rate R bps. Suppose that the two hosts are separated by m meters, and suppose the propagation speed along the link is s meters/sec. Host A is to send a packet of size L bits to Host B.

- Express the propagation delay, d_{prop} , in terms of m and s .
- Determine the transmission time of the packet, d_{trans} , in terms of L and R .
- Ignoring processing and queuing delays, obtain an expression for the end-to-end delay.
- Suppose Host A begins to transmit the packet at time $t = 0$. At time $t = d_{\text{trans}}$, where is the last bit of the packet?
- Suppose d_{prop} is greater than d_{trans} . At time $t = d_{\text{trans}}$, where is the first bit of the packet?
- Suppose d_{prop} is less than d_{trans} . At time $t = d_{\text{trans}}$, where is the first bit of the packet?
- Suppose $s = 2.5 \cdot 10^8$, $L = 120$ bits, and $R = 56$ kbps. Find the distance m so that d_{prop} equals d_{trans} .

- $d_{\text{prop}} = m/s$
- $d_{\text{trans}} = L/R$
- $d_{\text{end-end}} = m/s + L/R$
- Just left Host A.

(e) On the link.

(f) At Host B.

(g) $d_{\text{prop}} = m/s = d_{\text{trans}} = L/R$

$$m = s \cdot L/R = 2.5 \times 10^8 \cdot 120/56 \times 10^3 \approx 536 \times 10^3 m = 536 km$$

P10. Consider a packet of length L which begins at end system A and travels over three links to a destination end system. These three links are connected by two packet switches. Let d_i , s_i , and R_i denote the length, propagation speed, and the transmission rate of link i , for $i = 1, 2, 3$. The packet switch delays each packet by d_{proc} . Assuming no queuing delays, in terms of d_i , s_i , R_i , ($i = 1, 2, 3$), and L , what is the total end-to-end delay for the packet? Suppose now the packet is 1,500 bytes, the propagation speed on all three links is $2.5 \cdot 10^8$ m/s, the transmission rates of all three links are 2 Mbps, the packet switch processing delay is 3 msec, the length of the first link is 5,000 km, the length of the second link is 4,000 km, and the length of the last link is 1,000 km. For these values, what is the end-to-end delay?

$$d_{\text{end-end}} = \frac{L}{R_1} + \frac{d_1}{s_1} + d_{\text{proc}} + \frac{L}{R_2} + \frac{d_2}{s_2} + d_{\text{proc}} + \frac{L}{R_3} + \frac{d_3}{s_3}$$

$$d_{\text{end-end}} = 3 \cdot \frac{1500 \times 8}{2 \times 10^6} + \frac{(5000 + 4000 + 1000) \times 10^3}{2.5 \times 10^8} + 2 \times 0.003 = 0.064s = 64ms$$

P12. A packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, one other packet is halfway done being transmitted on this outbound link and four other packets are waiting to be transmitted. Packets are transmitted in order of arrival. Suppose all packets are 1,500 bytes and the link rate is 2 Mbps. What is the queuing delay for the packet? More generally, what is the queuing delay when all packets have length L , the transmission rate is R , x bits of the currently-being-transmitted packet have been transmitted, and n packets are already in the queue?

$$d_{\text{queue}} = \frac{1500 \times 8}{2 \times 10^6} \cdot \left(\frac{1}{2} + 4 \right) = 0.027s = 27ms$$

$$d_{\text{queue}} = \frac{n \cdot L + (L - x)}{R}$$

P13. (a) Suppose N packets arrive simultaneously to a link at which no packets are currently being transmitted or queued. Each packet is of length L and the link has transmission rate R . What is the average queuing delay for the N packets?

(b) Now suppose that N such packets arrive to the link every LN/R seconds. What is the average queuing delay of a packet?

(a)

$$\bar{d} = \frac{1}{N} \cdot [1 + 2 + \dots + (N - 1)] \cdot \frac{L}{R} = \frac{1}{N} \cdot \frac{N \cdot (N - 1)}{2} \cdot \frac{L}{R} = \frac{(N - 1) \cdot L}{2R}$$

(b) Same. It takes $N \cdot L/R$ to transmit a batch of packets, so the average queuing delay is the same for all batches.

P1 (15 Points) Consider the network in Fig. 1, where router 1, 2, 3 and 4 forward packets with rate $R_1=0.5$ Mbit/s, $R_2=2$ Mbit/s, $R_3=1$ Mbit/s and $R_4=1$ Mbit/s, respectively. Router 1 and 2 start the transmission of packet A and packet B simultaneously, and the buffer of router 3 and 4 are empty. The size of packet A and B is $L=1$ Mbit. (Hint: fixed deterministic rate!). Assume processing and propagation delay are negligible.

- Find the end-to-end delay of packet A.
- Solve the same when $R_3=0.5$ Mbit/s, with the rate of other routers unchanged.
- Solve it when $R_3=0.5$ Mbit/s and $R_4=0.25$ Mbit/s, with the rate of other routers unchanged.

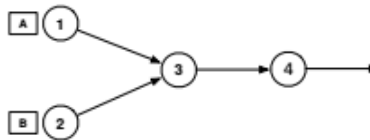


Fig. 1. Problem P1

- (a) 4 sec
- (b) 5.5 sec
- (c) 10.5 sec

P2 (15 Points) Consider a router whose transmission rate is R bit/s (fixed rate). The time between two consecutive packet arrival in the (infinite) buffer is an exponential variable with rate λ . The length (in bits) of the packets is an exponential variable with average L bits.

- assume a packet arrived at time t seconds, find the probability that 0 further new packets are arrived in the buffer at time $t+s$ seconds.
- Find the probability that the transmission time of a packet is larger than τ seconds.
- Assume that a new packet (packet A) arrives at time $t=0$, and that N packets are in the system (including the packet currently being served). Define T as the time when the last bit of packet A is transmitted. Find $E[T]$.

$$P(X > t) = \int_t^{\infty} \lambda e^{-\lambda x} dx = \lambda \frac{-e^{-\lambda x}}{\lambda} \Big|_t^{\infty} = e^{-\lambda t}.$$

- (a) According to the memoryless property of exponential distribution

$$\begin{aligned}
 P(Y > t | X > s) &= \frac{P(X > s+t | X > s)}{P(X > s+t, X > s)} \\
 &= \frac{P(X > s+t)}{P(X > s)} \\
 &= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} \\
 &= e^{-\lambda t}.
 \end{aligned}$$

The answer is $P = e^{-\lambda s}$

- (b) The length is larger than $R \cdot \tau$ if the transmission time is larger than τ . Hence, the probability is

$$P = e^{-R \cdot \tau / L}$$

- (c) The expected transmission time for one packet is L/R . Hence the answer is

$$\frac{L}{R}(N+1)$$

P3 (20 Points) A router receives packets from two applications. The time between the generation of consecutive packets of application 1 and application 2 are exponential variables with rate λ_1 and λ_2 , respectively. The service time of the router is an exponential variable with rate μ . Assume 1 packet (packet A) is in the buffer (and is being served) at time 0.

- Find the probability that a, randomly selected, incoming packet is generated from application 1.
- After t_B seconds, a new packet (packet B) arrives in the buffer. Find $E[t_B]$.
- Define N as the number of packets in the buffer after packet B arrives. Find $E[N]$.
- Compute the average permanence time of packet B in the buffer (waiting time plus transmission).
- Define T as the time when the last bit of the next packet arrived in the buffer is transmitted by the router. Find $E[T]$.

$$\frac{\lambda_1}{\lambda_1 + \lambda_2}$$

$$\frac{1}{\lambda_1 + \lambda_2}$$

Case	Probability	(c)	(d)	(e)
A leaves first	$\frac{\mu}{\lambda_1 + \lambda_2 + \mu}$	1	$\frac{1}{\mu}$	$\frac{1}{\mu} + \frac{1}{\lambda_1 + \lambda_2} + \frac{1}{\mu}$
B arrives first	$\frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + \mu}$	2	$\frac{2}{\mu}$	$\frac{2}{\mu}$
Sum	1			

$$\frac{\mu}{\lambda_1 + \lambda_2 + \mu} \cdot 1 + \frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + \mu} \cdot 2 = \frac{2 \cdot (\lambda_1 + \lambda_2) + \mu}{\lambda_1 + \lambda_2 + \mu}$$

$$\frac{\mu}{\lambda_1 + \lambda_2 + \mu} \cdot \frac{1}{\mu} + \frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + \mu} \cdot \frac{2}{\mu}$$

$$\frac{\mu}{\lambda_1 + \lambda_2 + \mu} \cdot \left(\frac{1}{\mu} + \frac{1}{\lambda_1 + \lambda_2} + \frac{1}{\mu} \right) + \frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + \mu} \cdot \frac{2}{\mu}$$

Solution 2 of this part is

$$\frac{1}{\lambda_1 + \lambda_2} + \frac{\mu}{\lambda_1 + \lambda_2 + \mu} \cdot \frac{1}{\mu} + \frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + \mu} \cdot \frac{2}{\mu}$$

Where the first term is the time it takes before packet B arrives. Since then, if A leaves first, then after B comes, it takes $1/\mu$ to transmit B. Otherwise, i.e. if B arrives first, then after B arrives, it takes $1/\mu$ to transmit A, then another $1/\mu$ to transmit B, which in total is $2/\mu$.

Solution 3 of this part is

$$\frac{1}{\lambda_1 + \lambda_2 + \mu} + \frac{\mu}{\lambda_1 + \lambda_2 + \mu} \cdot \left(\frac{1}{\lambda_1 + \lambda_2} + \frac{1}{\mu} \right) + \frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + \mu} \cdot \frac{2}{\mu}$$

Where the first term is the time it takes before the first event occurs, which could be A's transmission or B's arrival. If the first event is A's transmission, then it takes time to wait for B to come and then transmit it. If the first event is B's arrival, then it takes time to transmit both packets.